Predictive Customer Churn Analysis Using Logistic Regression in Orange Data Mining

To understand customer churn dynamics and predict potential subscriber loss, an end-to-end classification pipeline was constructed in **Orange Data Mining**. Using an operational dataset of **64,374 customer records**, key behavioral attributes such as support calls, payment delays, spend, and contract length were processed to train a **Logistic Regression** model. The analysis follows a structured stage-wise approach: beginning with feature selection and model training, progressing through rigorous validation using 10-fold stratified cross-validation, and concluding with real-world confusion matrix evaluation and instance-level risk scoring to drive targeted customer retention strategies.

1. Raw Dataset Overview

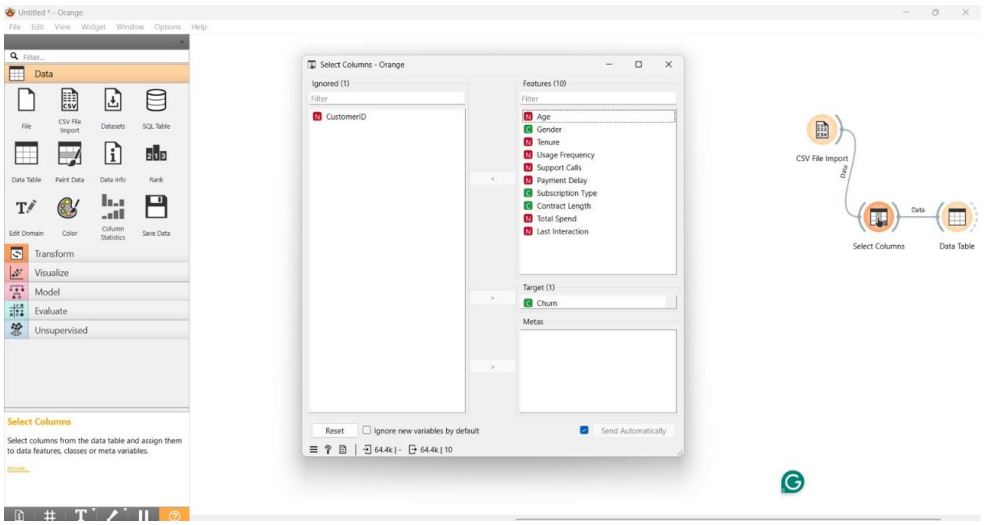
Data: customer_churn_dataset-testing-master: 64374 instances, 12 variables
Features: 12 (4 categorical, 8 numeric) (no missing values)

	Support Calls	Payment Delay	Subscription Type	Contract Length	Total Spend	Last Interaction	Churn
1		27	Basic	Monthly	598	9	1
2		13	Standard	Monthly	584	20	0
3		29	Premium	Annual	757	21	0
4		17	Premium	Quarterly	232	18	0
5		2	Standard	Annual	533	18	0
6		10	Premium	Monthly	500	29	0
7		28	Basic	Quarterly	574	14	1
8		18	Standard	Monthly	323	16	0
9		8	Basic	Monthly	687	8	0
10		23	Basic	Annual	995	10	0
11		21	Standard	Annual	526	3	1
12		14	Basic	Quarterly	187	1	0

This image displays the raw dataset uploaded into Orange, named **customer_churn dataset-testing-master**. The dataset contains **64,374 instances (rows)** and **12 variables (columns)**, divided into 4 categorical and 8 numerical features, with no missing values. Columns include operational metrics like Support Calls, Payment Delay, Subscription Type, Contract Length, Total Spend, Last Interaction, and the target outcome Churn.

The data represents customer activity and engagement records used to analyze churn behaviors. Since there are zero missing values across all 64,374 instances, the dataset is clean and ready for preprocessing without requiring imputation techniques. The target column Churn is binary (0 for retained, 1 for churned).

2. Feature Selection & Target Definition



This screenshot highlights the **Select Columns** widget configuration panel. The user has organized the dataset into three distinct functional categories:

- **Ignored:** CustomerID (removed because individual identification numbers carry no statistical predictive power).
- **Features (10):** Age, Gender, Tenure, Usage Frequency, Support Calls, Payment Delay, Subscription Type, Contract Length, Total Spend, and Last Interaction.
- **Target (1):** Churn (the class label to be predicted).

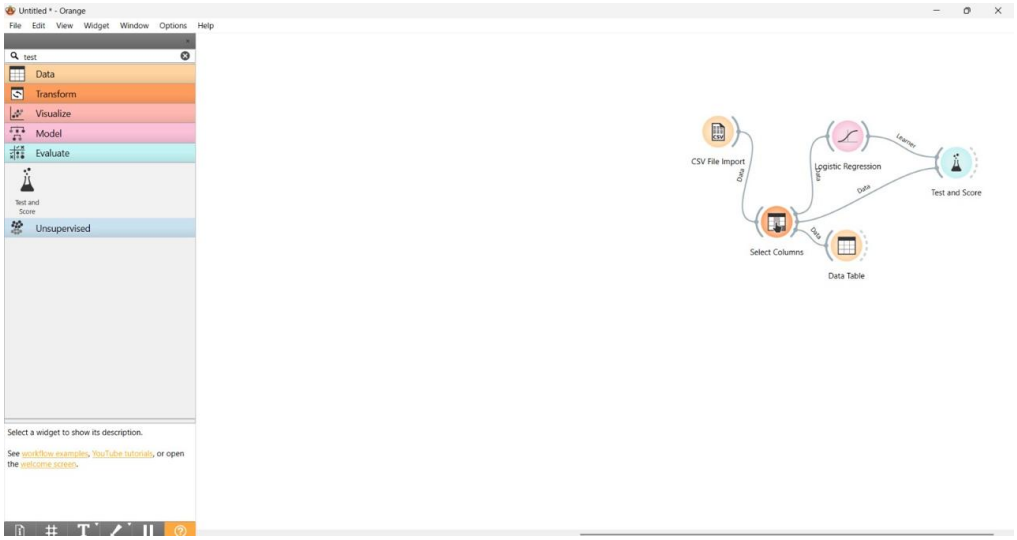
Explicitly removing non-predictive metadata like CustomerID prevents model overfitting and data noise. Setting Churn as the single target variable signals to Orange that this workflow will train a supervised classification model rather than an unsupervised clustering or regression algorithm.

3. Basic Machine Learning Workflow Construction



The data flows from loading (CSV File Import) to attribute mapping (Select Columns). The structured data is then simultaneously routed in two directions: into a tabular view (Data Table) for verification, and into the Logistic Regressionalgorithm to establish a baseline classification model.

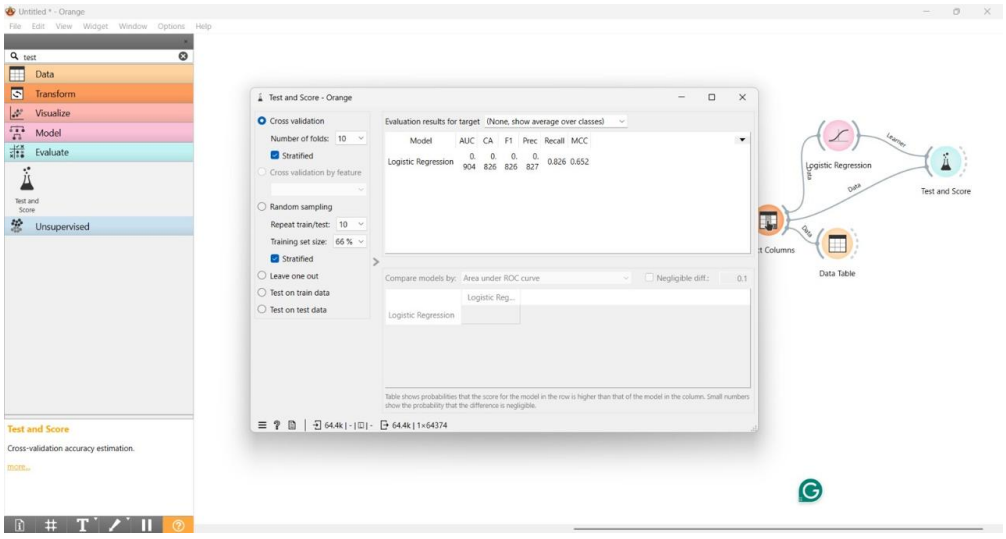
4. Cross-Validation Integration Setup



The canvas is expanded to incorporate the **Test and Score** widget. The output from Select Columns connects to Test and Scoreas raw data, while the Logistic Regression node connects as a **Learner**.

Connecting a learner and dataset to Test and Score allows the system to run validation procedures (like Stratified K-Fold cross-validation) rather than evaluating the model solely on its training data, preventing biased evaluation.

5. Stratified Cross-Validation Evaluation Results



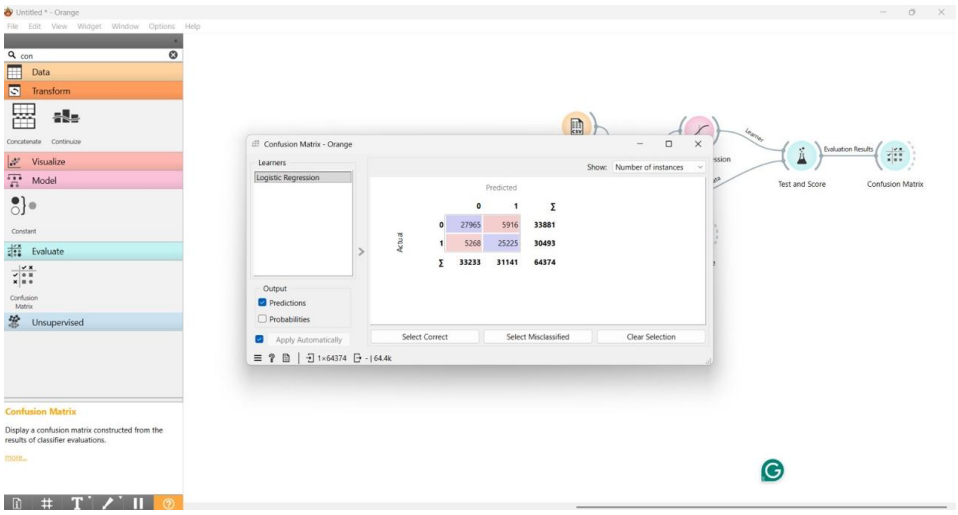
This window displays the validation scores computed inside the **Test and Score** widget using **10-fold Stratified Cross-Validation**.

- **AUC (Area Under ROC Curve):** 0.904
- **CA (Classification Accuracy):** 0.827 (82.7%)
- **F1 Score:** 0.826
- **Precision:** 0.827
- **Recall:** 0.827
- **MCC (Matthews Correlation Coefficient):** 0.652

Interpretation:

- An **AUC of 0.904** demonstrates excellent class separation capacity.
- An overall **Accuracy of 82.7%** combined with balanced **Precision** and **Recall** (82.7%) indicates the Logistic Regression model performs reliably across both churners and non-churners without heavy skew.
- An **MCC of 0.652** confirms a strong positive correlation between actual outcomes and predictions, far superior to random guessing (0.0).

6. Confusion Matrix Analysis



In the analysis of **64,374 total customers**, the confusion matrix compares what the **model predicted** against what **actually happened**:

	Predicted Retained (0)	**Predicted Churned (1)**	**Total Actual**
Actual Retained (0)	**27,965** *(True Negative)*	**5,916** *(False Positive)*	**33,881**
Actual Churned (1)	**5,268** *(False Negative)*	**25,225** *(True Positive)*	**30,493**
Total Predicted	**33,233**	**31,141**	**64,374**

Real-Life Business Interpretation

To understand what these numbers mean in the real world, let's look at each of the 4 quadrants:

1. True Negatives (27,965 Customers) "*The Happy Loyalists*"

- The model predicted these customers would stay, and they actually stayed. These are stable, satisfied customers. We don't need to waste marketing budget offering them discounts or intervention incentives.

2. True Positives (25,225 Customers) "*The Rescued At-Risk Customers*"

- The model predicted these customers were going to leave (churn), and they were indeed planning to leave. Because the model correctly flagged these 25,225 individuals, the retention team can proactively offer them a contract discount, a loyalty reward, or a quick support check-in to convince them to stay before they cancel.

3. False Positives (5,916 Customers) "*The Unnecessary Perks*"

- The model predicted these customers would leave, but they actually had no intention of leaving. This is a **False Alarm**. Sending these customers special retention discounts or freebies means spending money on people who were going to stay anyway. While it costs the company a bit of unnecessary retention budget, it doesn't harm customer trust.

4. False Negatives (5,268 Customers) "*The Silent Losses (Critical Risk)*"

- **What it means:** The model predicted these customers were safe and would stay, but they actually churned and left the service. **This is the most dangerous error for a business.** These 5,268 customers left completely under the radar. Because the model assumed they

were happy, no retention offer or customer service call was made, resulting in direct revenue loss.

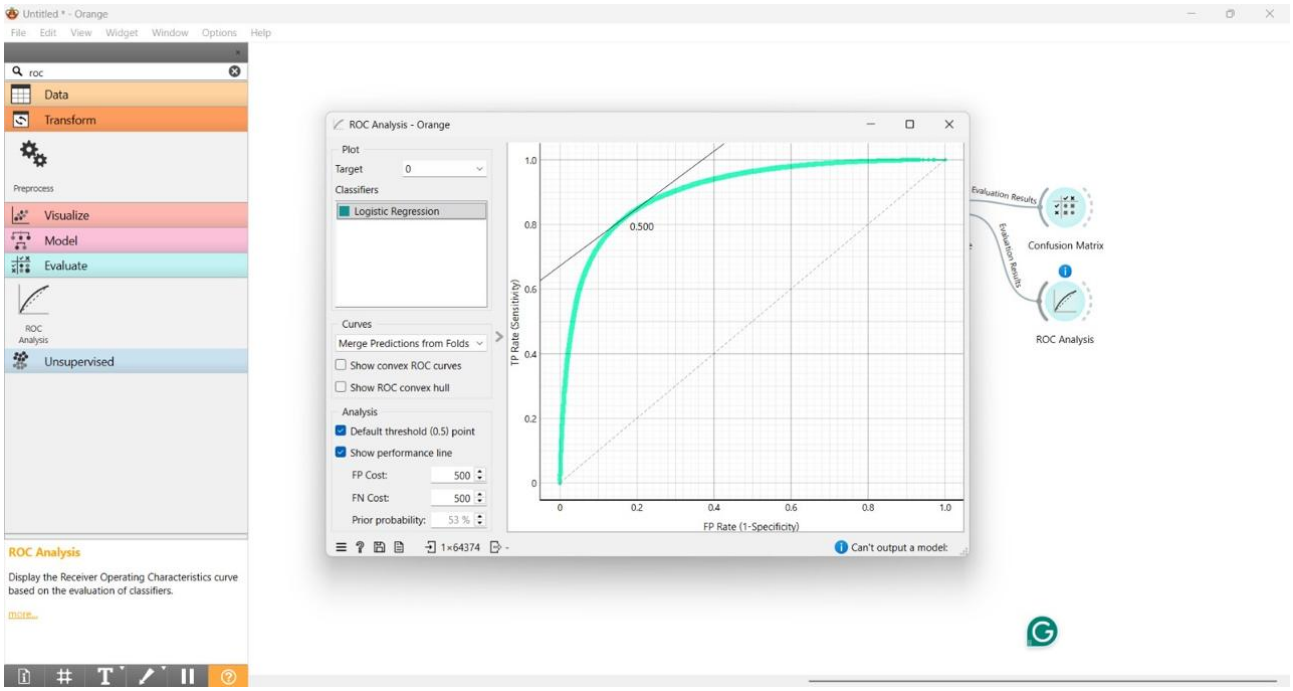
Key Business Takeaways from the Analysis

1. **Overall Accuracy (82.63%):** Out of 64,374 customers, the model got 53,190 predictions right (27,965+25,225). This gives a very solid baseline for automated churn prediction.

2. **Churn Detection Rate / Recall (82.7%):** The model successfully identified 25,225 out of 30,493 total churners (82.7%). This means the business can capture over 4 out of 5 at-risk customers before they walk away.

3. **Business Recommendation:** Because a **False Negative** (losing a customer forever) is almost always far more expensive than a **False Positive** (giving a minor discount to a loyal customer), the company can consider adjusting the model's classification threshold slightly lower (e.g., to 0.40 instead of 0.50) to catch even more of those 5,268hidden churners.

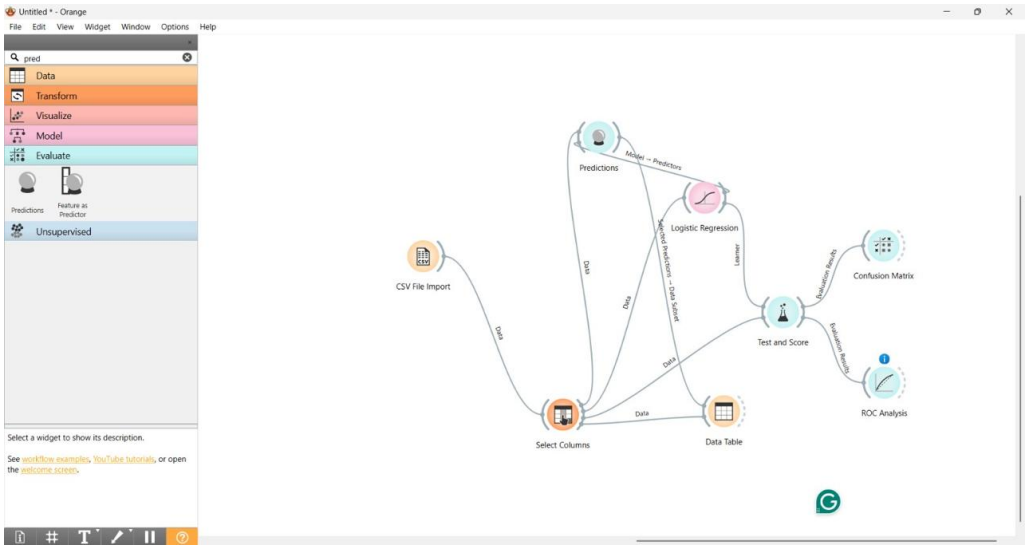
7. Receiver Operating Characteristic (ROC) Curve



This chart displays the **ROC Curve Analysis** (True Positive Rate vs. False Positive Rate) for Target Class 0. The cyan line traces performance across all probability thresholds, anchored by a cost tangent line at the default threshold point (0.500). The prior probability is noted at 53% (33,881/64,374).

The steep upward slope towards the top-left corner highlights high sensitivity accompanied by low false alarm rates. The black line tangent to the curve confirms that a decision threshold around 0.50 optimizes classification accuracy given symmetric misclassification costs.

8. Final Integrated Orange Workflow



This screenshot illustrates the complete, fully connected Orange workflow scheme. Data flows from CSV File Import to Select Columns, which simultaneously feeds:

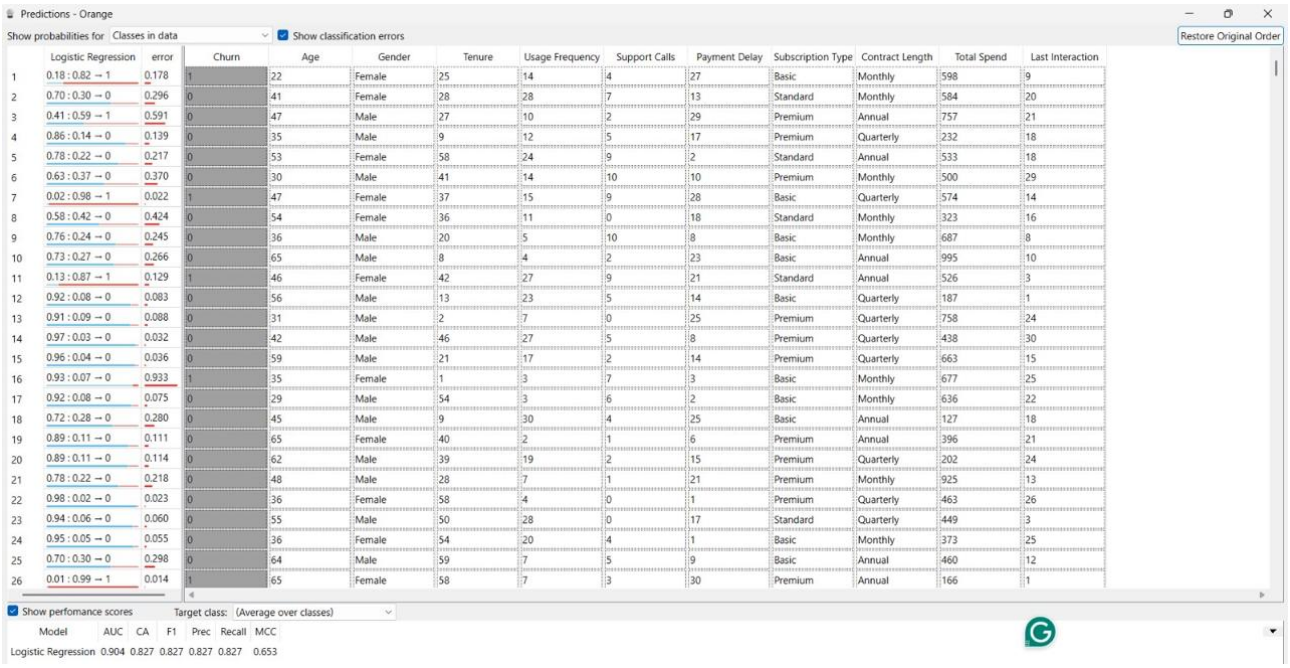
1. Data Table (data inspection)
2. Test and Score (evaluation pipeline, which outputs to Confusion Matrix and ROC Analysis)
3. Predictions (inference pipeline combining raw data and the trained Logistic Regression model)

This schema forms an end-to-end Machine Learning pipeline in Orange. It separates model training and cross-validation from inference generation, enabling real-time probability estimations on input dataset records.

9. Record-Level Predictions & Instance Output Table

This window translates abstract model performance into individual customer profiles. Each row represents a real customer, showing how their usage patterns convert into a churn risk score:

- **Logistic Regression (P(Stay):P(Churn) - Prediction):** The model's estimated probability distribution for the customer. For instance, 0.18 : 0.82 > 1 means the model calculates an



18% chance of staying and an **82% chance of churning**, predicting the final status as **1** (Churn).

- **Error:** Quantifies prediction uncertainty or loss value (1−P(True Class)). A smaller value means the model is highly confident in its correct prediction.
- **Churn (Target):** The actual ground-truth outcome (0 = Stayed, 1 = Left).
- **Customer Attributes (Features):** Key operational metrics such as Age, Tenure (months active), Support Calls, Payment Delay, Subscription Type, Contract Length, Total Spend, and Last Interaction.

Real-Life Scenario Analysis :

To see how customer support or account management teams would use this table daily, let's examine specific customer profiles from the image:

Scenario A: High-Risk Churner Correctly Flagged (Row 1)

- **Customer Profile:** A 22-year-old female with a basic monthly contract, 4 support calls, and a massive **27-day payment delay**.
- **Model Output:** P(Churn)=0.82, predicted as **1** (Actual = 1, Error = 0.178).
- **Real-World Meaning:** The model easily detected that a long payment delay paired with frequent support calls signal high customer frustration.
- **Actionable Step:** Trigger an automated high-priority retention workflow, offer a payment grace period or a temporary contract upgrade before the subscription cancels.

Scenario B: Unhappy Customer Caught Early (Row 7)

- **Customer Profile:** A 47-year-old female on a quarterly basic plan with 9 support calls and a **28-day payment delay**.
- **Model Output:** P(Churn)=0.98, predicted as **1** (Actual = 1, Error = 0.022).
- **Real-World Meaning:** With a 98% churn probability and near-zero error (0.022), this customer is practically out the door due to poor experience/unresolved issues.
- **Actionable Step:** Assign a dedicated customer success representative to personally reach out and resolve her open support tickets immediately.

Scenario C: Safe, Satisfied Customer (Row 2)

- **Customer Profile:** A 41-year-old female on a monthly standard plan with a **13-day payment delay** and low support friction.
- **Model Output:** P(Churn)=0.30, predicted as **0** (Actual = 0, Error = 0.296).
- **Real-World Meaning:** Even though there is a minor payment delay, her overall profile indicates low churn risk (70% chance of staying).
- **Actionable Step: Do nothing.** Avoid sending unnecessary promotional spam or aggressive discount offers that eat into profit margins.

Scenario D: False Alarm / False Positive (Row 6)

- **Customer Profile:** A 30-year-old male with a high support call volume (10 calls) and a 10-day payment delay.
- **Model Output:** P(Churn)=0.37 (Row 6: 0.63 : 0.37 > 0, Actual = 0).
- **Real-World Meaning:** Despite calling support frequently, his overall activity keeps him leaning toward staying (63% probability of staying). The model avoids prematurely declaring him a lost cause.

How Business Teams Operationalize This Table

Instead of looking at 64,374 rows manually, business managers set up automated action triggers based on probability thresholds:

1. **Urgent Tier (P(Churn)≥0.80):** High risk of leaving immediately. Send immediate automated intervention, account manager outreach, or steep renewal discounts.
2. **Watchlist Tier (0.50≤P(Churn)<0.80):** Moderately dissatisfied. Queue for automated email surveys, feature tutorials, or subtle retention prompts.
3. **Safe Zone (P(Churn)<0.50):** Low risk. Proceed with standard billing and operational communications without intervention costs.